American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

CN Lab Lecture 2 IP_Addressing II

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Semester: Summer 2025-2026 **Date: June 19, 2026**

Learning Objective Answers

1. **What is the difference between a Network ID and a Host ID?**

   The Network ID identifies the network to which a device belongs, whereas the Host ID identifies a specific device within that network. Together, they form a complete IPv4 address.

2. **How can you identify a Class B IP address?**

   A Class B IP address can be identified if its first two bits are `10` in binary or if its first octet falls within the range `128-191`.

3. **What is the purpose of a broadcast address?**

   A broadcast address is used to send a packet to all devices within a network simultaneously. It is the highest address in a network and has all host bits set to 1.

4. **What is the default subnet mask of a Class C network?**

   The default subnet mask of a Class C network is:

   $$255.255.255.0$$

   or

   $$/24$$

   which means the first 24 bits represent the Network ID and the remaining 8 bits represent the Host ID.

1. **Analyze the structure of classful IPv4 addressing and compare the addressing schemes of Class A, Class B, and Class C networks.**

   Classful IPv4 addressing divides the 32-bit IPv4 address into Network ID and Host ID portions. The size of these portions depends on the address class.

Class	First Octet Range	Network Bits	Host Bits	Default Mask
A	1–126	8	24	255.0.0.0
B	128–191	16	16	255.255.0.0
C	192–223	24	8	255.255.255.0

   Class A supports a small number of networks with a very large number of hosts. Class B provides a balance between networks and hosts. Class C supports a large number of networks but fewer hosts per network. Therefore, Class A is suitable for very large organizations, Class B for medium-sized organizations, and Class C for small organizations.

2. **Demonstrate how to identify the class of an IPv4 address using both binary notation and dotted-decimal notation. Provide suitable examples.**

   The class of an IPv4 address is determined by its leading bits or the value of its first octet.

Class	Leading Bits	First Octet Range
A	0	1–126
B	10	128–191
C	110	192–223
D	1110	224–239
E	1111	240–255

Examples:

- 10.5.3.5 → First octet = 10 → Class A
- 172.16.5.10 → First octet = 172 → Class B
- 192.168.1.1 → First octet = 192 → Class C

In binary, the first octet of 172 is:

$$172 = 10101100_2$$

Since it begins with `10`, it is a Class B address.

3. **Explain the role of subnet masks in classful addressing. Discuss the default subnet masks of different address classes and their significance.**

   A subnet mask is a 32-bit value used to separate the Network ID from the Host ID in an IP address. The bits set to 1 represent the network portion, while the bits set to 0 represent the host portion.

   Default subnet masks are:

Class	Default Mask	CIDR Notation
A	255.0.0.0	/8
B	255.255.0.0	/16
C	255.255.255.0	/24

   Subnet masks help routers and hosts determine whether a destination belongs to the same network or a different network. They are essential for network organization, routing, and address management.

4. **A host is assigned the IP address 172.10.12.10. Determine the network address, broadcast address, valid host range, and total number of usable hosts. Show all calculations.**

   Given IP address:

$$172.10.12.10$$

   Since the first octet is 172, it is a **Class B** address.

   Default subnet mask:

$$255.255.0.0\ (/16)$$

   Network ID = First two octets:

$$172.10$$

   Host ID = Last two octets:

$$12.10$$

   Network Address:
   Replace all host bits with 0:

$$172.10.0.0$$

   Broadcast Address:
   Replace all host bits with 1:

$$172.10.255.255$$

Valid Host Range:

$$172.10.0.1 \text{ to } 172.10.255.254$$

Total Addresses:

$$2^{16} = 65,536$$

Usable Hosts:

$$2^{16} - 2 = 65,534$$

Therefore,

- Network Address = 172.10.0.0
- Broadcast Address = 172.10.255.255
- Host Range = 172.10.0.1 – 172.10.255.254
- Usable Hosts = 65,534

5. **Explain how address blocks are analyzed in IPv4 networking. Discuss the methods for determining the total number of addresses, the first address, and the last address in a block.**

   An address block is a contiguous range of IP addresses assigned to a network. To analyze an address block, the following information is determined:

   (a) Total number of addresses.
   (b) First address (Network Address).
   (c) Last address (Broadcast Address).

   If the network contains n network bits, then:

   $$\text{Total Addresses} = 2^{32-n}$$

   The first address is obtained by setting all host bits to 0, while the last address is obtained by setting all host bits to 1.

   For example, consider the block:

   $$192.168.1.0/24$$

   Number of host bits:

   $$32 - 24 = 8$$

   Total addresses:

   $$2^8 = 256$$

   First address:

   $$192.168.1.0$$

   Last address:

   $$192.168.1.255$$

   Usable host range:

192.168.1.1 to 192.168.1.254

Thus, address block analysis helps network administrators determine the size of a network, available host addresses, and the corresponding network and broadcast addresses.

Explain the concept of broadcast communication in IPv4 networks. Differentiate between limited broadcast and direct broadcast with suitable examples.

Broadcast communication is a method of sending a packet from one host to all hosts within a network. Instead of sending separate packets to each device, a single broadcast packet is transmitted, and every host in the target network receives and processes it. Broadcast communication is commonly used for network discovery, address resolution, and service announcements.

There are two types of broadcast communication:

1. **Limited Broadcast**
   - A packet is sent to all hosts within the same network.
   - The destination IP address is `255.255.255.255`.
   - Routers do not forward limited broadcast packets.

   Example:

   If a host with IP address `192.168.1.10` wants to send a message to all devices on its local network, it sends the packet to:

   255.255.255.255

   All hosts on the local network receive the packet.

2. **Direct Broadcast**
   - A packet is sent to all hosts on a specific remote network.
   - The destination address is the broadcast address of the target network.
   - Routers may forward direct broadcasts if configured to do so.

   Example:

   Consider the Class A network:

   20.0.0.0

   The broadcast address of this network is:

   20.255.255.255

   A packet sent to `20.255.255.255` will be delivered to all hosts within the `20.0.0.0` network.

Comparison of Limited and Direct Broadcast

Feature	Limited Broadcast	Direct Broadcast
Destination Address	255.255.255.255	Network Broadcast Address
Scope	Local Network Only	Specific Remote Network
Router Forwarding	Not Allowed	May Be Allowed
Example	255.255.255.255	20.255.255.255